The Impact of Natural Disasters on Stock Market: Evidence from Japan

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May 26, 2024

Abstract

This study examines the impact of natural disasters on the Japanese stock market, focusing specifically on the Nikkei index and key sectors including insurance, transportation, and utilities. Employing ARMA-X and GARCH models, we assess the immediate (first order) and subsequent (second order) effects of these events. Our findings reveal that while natural disasters do not significantly alter stock price levels within the Japanese market, they do lead to significant increases in market volatility. The transportation sector experiences the highest increase in volatility, while the impact on the Nikkei index, utilities, and insurance sectors is more moderate.

Keywords: Japan, ARMA-X, GARCH, Natural disasters, Nikkei

1 Introduction

Japan is among the countries that are the most affected by natural disasters due to its geographic location. It lies at the junction of the Pacific Plate and the Eurasian Plate, where the interactions between these two plates frequently cause earthquakes and tsunamis. This vulnerability has caused significant damage to the Japanese economy. A notable example is the Fukushima nuclear accident on March 11, 2011, which was caused by a massive tsunami following a magnitude 9 undersea earthquake (Kim et al., 2013). This disaster was extraordinary in terms of its economic damages.

Natural disasters can significantly impact Japan's financial markets due to its economic size and vulnerability to events like earthquakes and typhoons. As one of the largest global financial markets, second only to the U.S. (Fakhry et al., 2018), understanding Japan's market reactions to such disasters is crucial.

Understanding financial market fluctuations involves analyzing the psychological and behavioral responses of investors to new information, rather than just the information itself. This concept is particularly important in the context of natural disasters, as investor sentiment and behavioral biases play significant roles in market dynamics (Fakhry, 2016). Analyzing how market participants react to such events helps in understanding the resultant volatility and market efficiency.

2 Data

2.1 Financial indicators

We compiled extensive datasets that cover a wide range of relevant economic and financial indicators specific to Japan. Our analysis covers major stock indices and specific economic sectors, highlighting various economic impacts of natural disasters. These datasets have been collected from Yahoo Finance.

We incorporated a first dataset, which provides daily closing prices of the Nikkei 225 stock index. This index serves as a barometer for the health and performance of the Japanese stock market, offering insights into the market reactions to disruptive events.

Our research further explores the impacts on specific economic sectors by examining three crucial sectors. We analyzed the TOPIX Transportation, TOPIX Electric Power and Gas, which we have renamed as Utilities, and TOPIX Financials (Ex Banks), which we have renamed as Insurance, due to its predominant composition of insurance companies.

The insurance sector is often directly affected by disaster-related claims and liabilities, transportation by operational disruptions, and utilities by infrastructure damages, making these datasets useful for a nuanced analysis of sectoral responses to natural disasters.

2.2 Natural disasters

Data concerning natural disasters were obtained from the Emergency Events Database (EM-DAT), a reputable source for comprehensive disaster-related statistics.

To precisely link the financial data with the occurrence of natural disasters, each dataset was enhanced with additional columns. Dummy variables were introduced to indicate the presence of a natural disaster on a particular day, with a value of 1 marking the occurrence of an event.

Furthermore, the datasets include columns that quantify the total economic damage caused by the disasters, expressed in thousands of dollars ('000). This metric is crucial for assessing the financial impact and establishing correlations with market performance.

3 Methodology

3.1 ARMA-X model

To assess the effects of natural disasters on the price levels of our indices, we used ARMA-X models. These models combine the autoregressive (AR) and moving-average (MA) components of ARMA models with the ability to investigate the effects of exogenous variables X_t on the dependent variable Y_t.

In this context, Y_t is the adjusted closing price of one of our indices, and X_t represents the occurrence of a natural disaster, either as a dummy variable or as economic damage. The assumption that X_t is strictly exogenous to Y_t is verified, as the closing prices of the Nikkei or sector indices do not influence the occurrence or severity of natural disasters.

Based on the AIC criterion, the selected ARIMA models are ARIMA (2,0,2) for the Nikkei Index, ARIMA (0,0,0) for the transportation sector, ARIMA (0,0,1) for the utilities sector, and ARIMA (0,0,0) for the insurance sector.

This results in the following equation for the Nikkei index:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \beta X_t + \epsilon_t \tag{1}$$

where Y_t is the closing price for the Nikkei index, ϕ_1 and ϕ_2 are the coefficients of the AR terms, θ_1 and θ_2 are the coefficients of the MA terms, β is the coefficient of the exogenous regressor (X_t), and ϵ_t is the error term.

Finally, we applied these models using dummy, damage, and log damage variables at various points in the damage distribution (all, median, 0.75 quantile, and 0.90 quantile) to decompose the effect and increase the likelihood of observing an effect.

3.2 GARCH model

In this analysis, we used the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model to assess the impact of natural disasters on the volatility of the Japanese stock market. A GARCH model is a statistical model that estimates the volatility of returns by incorporating past variances and past forecast errors, making it particularly useful for financial time series data where volatility clustering is common.

The GARCH (1,1) model, which we identified as the best model based on Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), was employed. To quantify the impact of natural disasters, we included dummy variables representing different severity levels of natural disasters as external regressors in our GARCH model. The model specification is:

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \gamma D_t \tag{2}$$

where σ_t^2 is the conditional variance at time t, ω is a constant, α measures the impact of past squared shocks, β captures the persistence of volatility from one period to the next and the coefficient of interest γ quantifies how much the conditional variance is affected by the occurrence of a natural disaster.

We evaluated the significance of these dummy variables by examining the coefficient γ and its p-value. When the dummy variable's coefficient was statistically significant, we computed the increase in volatility due to natural disasters by comparing the average conditional variances with and without the occurrence of a severe natural disaster. The increase in volatility was then expressed as the ratio of standard deviations.

3.3 Dynamic multiplier analysis

In our dynamic multiplier analysis, we assessed how shocks, such as natural disasters, propagate through time and affect the conditional volatility of different sectors.

Using the GARCH model, we simulated the response of conditional volatility to a shock. The initial variance was set based on the model parameters. We introduced a shock at a specific period by setting a dummy variable to indicate the occurrence of a natural disaster. The conditional variances were then simulated over multiple periods to observe the dynamic response. We computed the conditional volatility as the square root of the simulated variances and plotted the dynamic response over time. This analysis helps us understand the persistence and decay patterns of volatility following a shock.

4 Results

The ARMA-X model analysis on Nikkei data reveals that the coefficients for the dummy variables and damage variables, reflecting the impact of natural disasters, are statistically insignificant. This insignificance is evident from the high p-values across the board, which suggests that neither the occurrence of natural disasters (represented by the dummy variables) nor the scale of economic damage caused by these disasters (represented by the damage variables) significantly influence the log-differenced Nikkei data. Notably, even when natural disasters were categorized based on their damage severity—50th, 75th, and 90th percentiles—there was no discernible impact on the stock market fluctuations. These findings imply that the Japanese stock market, as represented by the Nikkei index, may be resilient or indifferent to the immediate effects of natural disasters, at least over the periods analyzed in this study. Similarly, the analysis of the three sectors—transportation, utilities, and insurance—also showed insignificant results, further underscoring the robustness of the stock market against natural disaster shocks in terms of price levels.

Table 1: ARMA-X Analysis on Nikkei - Dummies

	Dummy	Dummy 0.5	Dummy 0.75	Dummy 0.90
Coefficient	0.0009	0.0021	0.0025	-0.0003
P-value	0.6171097 Not significant	0.4193283 Not significant	0.4992958 Not significant	0.9556994 Not significant
Increase in levels				

Table 2: ARMA-X Analysis on Nikkei - Damages

	Damage	Damage 0.5	Damage 0.75	Damage 0.90
Coefficient	0.0001	0.0001	0.0001	-0.0001
P-value	0.3173868 Not significant	0.6171097 Not significant	0.6171097 Not significant	0.7389047 Not significant
Increase in levels				

The results from the GARCH analysis show significant variability in the impact of natural disasters on the volatility across different sectors. The Transportation sector experiences the highest increase in volatility, with up to a 153% rise for the 0.90 dummy variable threshold. The Utilities sector also shows a notable increase, reaching up to 64%. The Insurance sector, while significantly affected, exhibits a more moderate increase in volatility, with a maximum rise of 55%. The Nikkei index, representing the overall market, demonstrates the least sensitivity, with increases of 23% and 35% for the 0.75 and 0.90 dummy thresholds, respectively.

To further understand these differences, we compared the volatility of the dummy over the 0.75 threshold. The Transportation sector has the lowest baseline volatility when no natural disaster occurs. However, when a natural disaster over the 0.75 threshold happens, its volatility jumps by 96%, making it the most volatile sector. This indicates that natural disasters have a pronounced effect on the volatility of the Transportation sector, while their impact on other sectors, though significant, is more moderate.

The explanation could be that natural disasters can cause substantial damage to transportation infrastructure, halting operations, increasing repair costs, and leading to significant losses. In contrast, the Utilities sector benefits from regulated pricing that provides stable revenue streams, the Insurance sector has mechanisms to manage risk, and the Nikkei is well-diversified across multiple sectors.

Table 3: GARCH Analysis on Nikkei

	Dummy	Dummy 0.5	Dummy 0.75	Dummy 0.90
Coefficient	0.000011	0.000040	0.000092	0.000148
P-value	0.380333 Not significant	0.078337 Not significant	0.033583 Significant	0.053175 Significant
Increase in volatility			23%	35%

Table 4: GARCH Analysis on Transportation

	Dummy	Dummy 0.5	Dummy 0.75	Dummy 0.90
Coefficient	0.000062	0.000181	0.000476	0.000895
P-value	0.004202 Significant	0.001725 Significant	0.000097 Significant	0.003390 Significant
Increase in volatility	17%	45%	96%	153%

The dynamic multipliers represent how the conditional volatility of each sector's index responds to a shock at the 0.75 quantile of the damage distribution. The beta coefficient (β_1) from each GARCH model represents the persistence of volatility over time, indicating how quickly or slowly volatility returns to its baseline level after a shock. This coefficient is derived from the GARCH model's estimation of the conditional variance equation, where

Table 5: GARCH Analysis on Utilities

	Dummy	Dummy 0.5	Dummy 0.75	Dummy 0.90
Coefficient	0.000088	0.000136	0.000215	0.000378
P-value	0.000008 Significant	0.000005 Significant	0.000015 Significant	0.000067 Significant
Increase in volatility	18%	26%	40%	64%

Table 6: GARCH Analysis on Insurance

	Dummy	Dummy 0.5	Dummy 0.75	Dummy 0.90
Coefficient	0.000048	0.000078	0.000121	0.000409
P-value	0.013827 Significant	0.009839 Significant	0.023323 Significant	0.002317 Significant
Increase in volatility	7%	12%	19%	55%

a higher β_1 value implies greater persistence in volatility.

In our analysis, the transportation sector is the most affected by the shock but has the lowest persistence of volatility ($\beta_1 = 0.7910667$) and takes about 60 periods to stabilize. In contrast, the utilities sector experiences a less pronounced initial increase in volatility but exhibits greater persistence ($\beta_1 = 0.8552864$), taking around 150 periods to return to baseline levels. The insurance sector, with the highest beta coefficient ($\beta_1 = 0.8801025$), is the least affected initially but shows the most prolonged volatility, also taking about 150 periods to stabilize.

These differences indicate varying resilience and persistence in response to shocks across sectors. The transportation sector shows more resilience, the utilities sector demonstrates moderate persistence, and the insurance sector, despite a lower initial impact, has the highest persistence, suggesting long-term effects of even minor shocks.

While there is no direct literature to support these findings, several interpretations are possible: The transportation sector exhibits less persistence due to its uncertain environment, marked by changes in oil prices (Aggarwal et al., 2012), which increases its flexibility to absorb shocks. The utilities sector may be less flexible because regulatory constraints slow its response to shocks. The high persistence observed in the insurance sector may stem from the substantial claims following catastrophic events, leading to prolonged financial instability and necessitating significant adjustments.

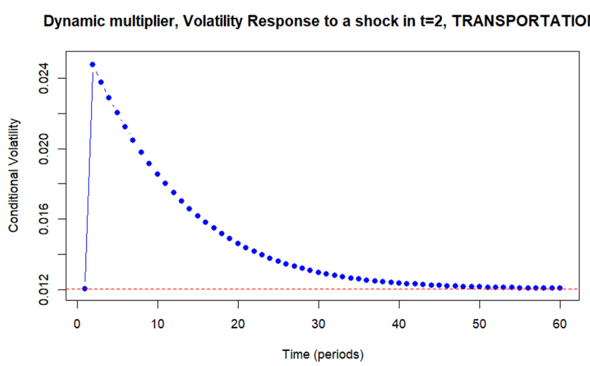

Figure 1: Dynamic multiplier Transportation

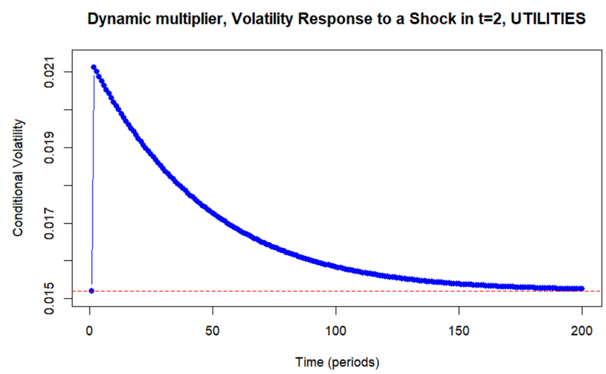

Figure 2: Dynamic multiplier Utilities

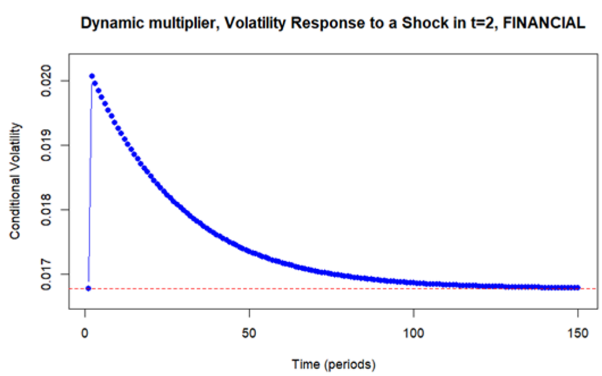

Figure 3: Dynamic multiplier Insurance

5 Conclusion

Our analysis of the impact of natural disasters on Japan's financial markets reveals distinct effects on both mean returns and volatility. The ARMA-X model analysis for the Nikkei index, as well as the Transportation, Utilities, and Insurance sectors, shows that natural disasters do not significantly affect mean returns. This is evident from the statistically insignificant coefficients for dummy and damage variables. In contrast, the GARCH analysis indicates that natural disasters significantly impact market volatility. The Transportation sector, in particular, experiences the highest increase in volatility, highlighting its vulnerability to infrastructure damage and operational disruptions. The Utilities and Insurance sectors exhibit more moderate increases in volatility, likely due to regulated pricing and effective risk management mechanisms.

Future research should expand this analysis to examine the impact of natural disasters in other parts of the world and their effects on Japan's financial markets. Additionally, by focusing on natural disasters occurring in specific areas of Japan and analyzing the stock returns of companies located in these regions, we could potentially observe significant effects on mean returns using the ARMA-X model.

Overall, our findings imply that natural disasters do not impact mean returns but significantly affect volatility. This information is crucial for guiding investors and policymakers in risk management and strategic planning.

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